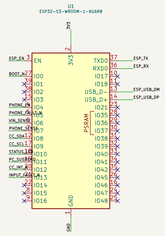
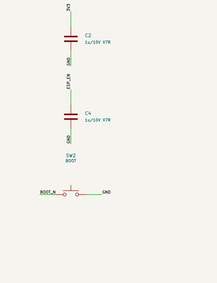
PRIVATE — A1 PCB REVIEW — NO MANUFACTURING RELEASE

01 ESP32-S3 / reset / status

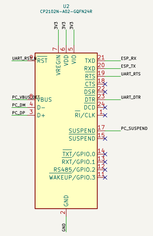




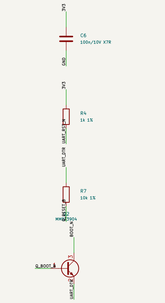


NC on unused GPIO is intentional. IO35/36/37 reserved for module PSRAM.

02 PC USB UART / auto download

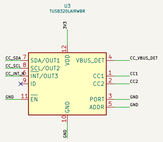








03 Type-C sink / power sensing



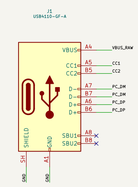




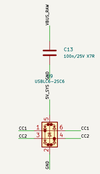


U3 PORT=GND: UFP; ADDR=GND: 0x47; EN_N=GND: internal Rd only.

04 PC USB-C / ESD / input

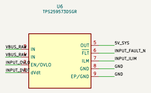








05 Input eFuse / controlled inrush

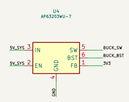








06 3.3V switching supply









AUTO RESET: DTR/RTS 00 or 11 -> EN/BOOT both high; 10 -> reset; 01 -> boot low.

07 Phone power / default OFF



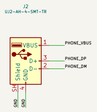






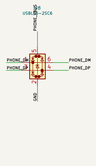
Firmware must verify >=1.5A source before PHONE_EN. 44.2k limit approximately 0.53–0.66A.

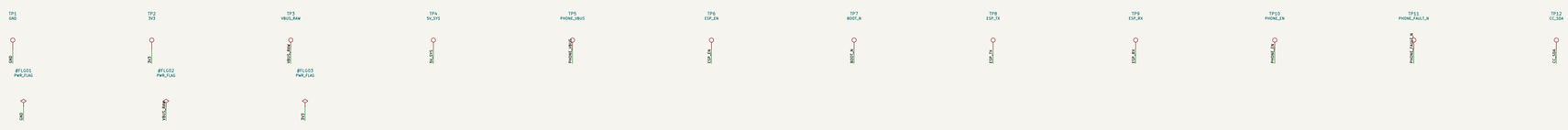
08 Android USB-A / USB FS











5V input only: phone VBUS defaults OFF; full SMT PROTOTYPE REVIEW — NOT MANUFACTURING RELEASE **Private project**		
Sheet: / File: phone-mcp-s3.kicad_sch		
Title: Phone MCP S3 — integrated USB controller		
Size: A1	Date: 2026-11-08	**Rev: A1 — routed PCB review**
KiCad E.D.A. 10.0.6		Id: 1/1

M1 NPTH 2.2mm M2 NPTH 2.2mm M3 NPTH 2.2mm M4 NPTH 2.2mm